Action: Special Relativity, Electromagnetism, and General Relativity

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1 Special Relativity

1.1 Lorentz Transformations

The Lorentz transformations relate coordinates between inertial reference frames:

$$x' = \gamma(x - vt) \tag{1}$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right) \tag{2}$$

where the Lorentz factor is:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{3}$$

1.2 Spacetime Interval and Metric

The proper time interval is defined as:

$$c^2 d\tau^2 = -ds^2 \tag{4}$$

The Minkowski metric (signature convention):

$$\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1) \tag{5}$$

1.3 Four-Vectors

1.3.1 Position Four-Vector

$$\bar{x} = (ct, x, y, z) = (x^0, x^1, x^2, x^3) \tag{6}$$

1.3.2 Four-Velocity

$$\bar{u} = \frac{d\bar{x}}{d\tau} = \gamma(c, \vec{v}) \tag{7}$$

1.3.3 Four-Momentum

$$\bar{p} = m_0\bar{u} = \left(\frac{E}{c}, \gamma m_0 \vec{v}\right) \quad \Rightarrow \quad p^0 = \frac{E}{c} \tag{8}$$

1.3.4 Four-Force

$$\bar{F} = \frac{d\bar{p}}{d\tau} \tag{9}$$

2 Electromagnetism in Special Relativity

2.1 Four-Vectors in Electromagnetism

2.1.1 Four-Derivative

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t}, \frac{\partial}{\partial x^i}\right) \tag{10}$$

This is a Lorentz covariant derivative operator.

2.1.2 Four-Current

$$J^\mu = (c\rho, \vec{J}) \tag{11}$$

The continuity equation:

$$\partial_\mu J^\mu = 0 \tag{12}$$

2.1.3 Four-Potential

$$A^\mu = \left(\frac{\phi}{c}, \vec{A}\right) \tag{13}$$

This transforms the same way as the position four-vector $\bar{x}$.

2.2 Electromagnetic Field Tensor

The electromagnetic field tensor is an antisymmetric rank-2 tensor:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix} \tag{14}$$

This is also called the Faraday tensor and satisfies:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{15}$$

2.3 Lorentz Force

The relativistic Lorentz force in covariant form:

$$f^\mu = qF^{\mu\nu}u_\nu \tag{16}$$

This generalizes the familiar $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$.
For non-relativistic velocities where $u \approx (1, \vec{v})$:

$$\frac{dp^i}{dt} = qE^i \quad \Rightarrow \quad F_{0i} \propto E_i \tag{17}$$

2.4 Field Transformations

Under Lorentz transformation $F' = \Lambda F \Lambda^T$, the electric and magnetic fields transform as:

Parallel components (unchanged):

$$E'_x = E_x \tag{18}$$

$$B'_x = B_x \tag{19}$$

Perpendicular components:

$$E'_y = \gamma(E_y - vB_z) \qquad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right) \tag{20}$$

$$E'_z = \gamma(E_z + vB_y) \qquad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right) \tag{21}$$

2.5 Maxwell's Equations in Covariant Form

Maxwell's equations can be written compactly as:

$$\boxed{\partial_\mu F^{\mu\nu} = \mu_0 J^\nu} \tag{22}$$

This encodes both Gauss's law and Ampère-Maxwell law.

2.6 Gauge Invariance

The electromagnetic action is invariant under gauge transformations:

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda(x) \tag{23}$$

where $\Lambda(x)$ is an arbitrary scalar function.

2.7 Action for Electromagnetism

The unique action that is both gauge and Lorentz invariant, and produces Maxwell's equations:

$$S_{EM} = \int d^4x \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu \right) \tag{24}$$

The Lagrangian density is:

$$\mathcal{L}_{EM} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu \tag{25}$$

3 General Relativity Foundations

3.1 Motivation for Curved Spacetime

In General Relativity, the motion of particles under gravity is determined by the metric $g_{\mu\nu}$. To find the metric for a given observer, we need field equations for $g_{\mu\nu}$.

Maxwell's theory of electromagnetism served as the guiding principle, as there is no direct Newtonian analog for the metric tensor.

3.2 Invariant Volume Element

3.2.1 Derivation

Consider a volume element transformation. We demand that the volume element be invariant:

$$dV = dV' \tag{26}$$

In terms of the Levi-Civita symbol:

$$J(\bar{x}) \frac{1}{2!} \epsilon_{\alpha\beta} dx^\alpha dx^\beta = J(\bar{x}') \frac{1}{2!} \epsilon_{\mu\nu} dx'^\mu dx'^\nu \tag{27}$$

where the permutation factor $(-1)^P$ accounts for cyclic ($P = 0$) and anti-cyclic ($P = 1$) permutations.

Under coordinate transformation $dx'^\mu = \Lambda^\mu_\alpha dx^\alpha$:

$$J(\bar{x}) = \det(\Lambda^\mu_\alpha) \tag{28}$$

This is the Jacobian of the coordinate transformation.

3.2.2 Connection to the Metric

The metric transforms as:

$$g'_{\mu\nu} = (\Lambda^{-1})^\alpha_\mu (\Lambda^{-1})^\beta_\nu g_{\alpha\beta} \tag{29}$$

Taking determinants:

$$\det\big(g'\big) = (\det \Lambda)^{-2} \det(g) \tag{30}$$

For Minkowski space $\eta_{\mu\nu}$ with $\det(\eta) = -1$:

$$\det(g) = (\det \Lambda)^2 \det(\eta_{\mu\nu}) \tag{31}$$

3.2.3 General Volume Element

The invariant volume element in D dimensions is:

$$\boxed{dV = \sqrt{|g|}\, d^D x} \tag{32}$$

where $g = \det(g_{\mu\nu})$ and we use $\sqrt{-g}$ for Lorentzian signatures.
Examples:

- **2D Polar Coordinates:** $(x, y) \to (r, \theta)$

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad \sqrt{|g|} = r, \quad dV = r\, dr\, d\theta \tag{33}$$

- **3D Spherical Coordinates:**

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\, d\phi^2 \tag{34}$$

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2\theta \end{pmatrix}, \quad \sqrt{|g|} = r^2 \sin\theta \tag{35}$$

$$dV = r^2 \sin\theta\, dr\, d\theta\, d\phi \tag{36}$$

3.3 Levi-Civita Symbol

The totally antisymmetric Levi-Civita symbol is defined as:

$$\epsilon_{ijk} = +1 \quad \text{for cyclic permutations of } (1, 2, 3) \tag{37}$$

$$\epsilon_{ijk} = -1 \quad \text{for anti-cyclic permutations} \tag{38}$$

$$\epsilon_{ijk} = 0 \quad \text{if any two indices are equal} \tag{39}$$

Application to determinants:
For a 2×2 matrix:

$$\det(A) = \frac{1}{2!} \epsilon_{\mu\nu} \epsilon_{\alpha\beta} a^{\mu\alpha} a^{\nu\beta} \tag{40}$$

For an $n \times n$ matrix:

$$\det(A) = \frac{1}{n!} \epsilon_{\mu_1 \cdots \mu_n} \epsilon_{\alpha_1 \cdots \alpha_n} a^{\mu_1 \alpha_1} \cdots a^{\mu_n \alpha_n} \tag{41}$$

4 Covariant Derivatives

4.1 Motivation

Consider whether $\partial_\mu g_{\alpha\beta}$ transforms as a tensor.
For a scalar field ϕ:

$$\phi(x) \to \phi'(x') = \phi(x) \tag{42}$$

The derivative of a scalar *is* a tensor:

$$\partial'_\mu \phi'(x') = (\Lambda^{-1})^\alpha_\mu \partial_\alpha \phi(x) \tag{43}$$

Therefore $\partial_\mu \phi(x)$ is a rank (0,1) tensor.
However, for a contravariant vector V^ρ with $V'^\rho = \Lambda^\rho_\beta V^\beta$:

$$\partial_\mu (V'^\rho) = (\Lambda^{-1})^\alpha_\mu \partial_\alpha (\Lambda^\rho_\beta V^\beta) \tag{44}$$

$$= (\Lambda^{-1})^\alpha_\mu \Lambda^\rho_\beta (\partial_\alpha V^\beta) + (\Lambda^{-1})^\alpha_\mu (\partial_\alpha \Lambda^\rho_\beta) V^\beta \tag{45}$$

Since $\Lambda^\rho_\beta = \Lambda^\rho_\beta(x)$ in general coordinate transformations, the second term does not vanish, so $\partial_\mu V^\rho$ does **not** transform as a rank (1,1) tensor.

We need a new derivative operator ∇_μ such that $\nabla_\mu V^\rho$ transforms properly as a tensor.

4.2 Definition of Covariant Derivative

We demand that:

$$\nabla'_\nu V'^\rho = (\Lambda^{-1})^\alpha_\nu \Lambda^\rho_\beta (\nabla_\alpha V^\beta) \tag{46}$$

In a locally inertial frame, the covariant derivative reduces to the ordinary derivative:

$$\nabla_\mu = \partial_\mu \quad \text{(in locally inertial frame)} \tag{47}$$

Through careful analysis of the transformation properties, we define:

$$\boxed{\nabla_\alpha V^\rho = \partial_\alpha V^\rho + \Gamma^\rho_{\alpha\beta} V^\beta} \tag{48}$$

where $\Gamma^\rho_{\alpha\beta}$ are the **Christoffel symbols** or **connection coefficients**.
For covariant vectors:

$$\boxed{\nabla_\alpha V_\beta = \partial_\alpha V_\beta - \Gamma^\rho_{\alpha\beta} V_\rho} \tag{49}$$

For scalars:

$$\nabla_\mu \phi = \partial_\mu \phi \tag{50}$$

For a mixed tensor T^α_β:

$$\nabla_\mu T^\alpha_\beta = \partial_\mu T^\alpha_\beta + \Gamma^\alpha_{\mu\rho} T^\rho_\beta - \Gamma^\rho_{\mu\beta} T^\alpha_\rho \tag{51}$$

General Rule: The covariant derivative transforms a tensor of rank (p, q) to a tensor of rank $(p, q+1)$.

4.3 Metric Compatibility

A fundamental requirement is that the metric is covariantly conserved:

$$\boxed{\nabla_\mu g_{\alpha\beta} = 0} \tag{52}$$

Substituting the definition:

$$\nabla_\mu g_{\alpha\beta} = \partial_\mu g_{\alpha\beta} - \Gamma^\rho_{\mu\alpha} g_{\rho\beta} - \Gamma^\rho_{\mu\beta} g_{\alpha\rho} = 0 \tag{53}$$

By cyclic permutation of indices and solving, we obtain:

$$\boxed{\Gamma^\rho_{\mu\alpha} = \frac{1}{2} g^{\rho\sigma} \left[\partial_\mu g_{\sigma\alpha} + \partial_\alpha g_{\mu\sigma} - \partial_\sigma g_{\mu\alpha}\right]} \tag{54}$$

This is the fundamental formula for Christoffel symbols in terms of metric derivatives.

4.4 Useful Identities

$$\Gamma^\alpha_{\mu\alpha} = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g}) \tag{55}$$

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} V^\mu) \quad \text{(covariant divergence)} \tag{56}$$

4.5 Leibniz Rule

The covariant derivative satisfies the Leibniz (product) rule:

$$\nabla_\mu (\phi V^\nu) = (\nabla_\mu \phi) V^\nu + \phi \nabla_\mu V^\nu \tag{57}$$

$$\nabla_\mu (V^\alpha W_\beta) = (\nabla_\mu V^\alpha) W_\beta + V^\alpha (\nabla_\mu W_\beta) \tag{58}$$

5 Curvature: The Riemann Tensor

5.1 Commutator of Covariant Derivatives

For any vector V^ρ, consider the commutator:

$$[\nabla_\mu, \nabla_\nu]V^\rho = \nabla_\mu(\nabla_\nu V^\rho) - \nabla_\nu(\nabla_\mu V^\rho) \tag{59}$$

After lengthy calculation:

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma \tag{60}$$

where the **Riemann curvature tensor** is defined as:

$$\boxed{R^\rho_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}} \tag{61}$$

This tensor contains at most two derivatives of the metric and measures the curvature of spacetime.

5.2 Properties of the Riemann Tensor

Property 1 (Antisymmetry in last two indices)**.**

$$R^\rho_{\sigma\mu\nu} = -R^\rho_{\sigma\nu\mu} \tag{62}$$

Property 2 (Antisymmetry in first pair)**.** *With all indices lowered:* $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^\lambda_{\sigma\mu\nu}$

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} \tag{63}$$

Proof of Property 2: Using metric compatibility:

$$0 = [\nabla_\mu, \nabla_\nu]g_{\alpha\beta} = [\nabla_\mu, \nabla_\nu](V_\alpha W_\beta) \tag{64}$$

This leads to:

$$R_{\alpha\beta\mu\nu} + R_{\beta\alpha\mu\nu} = 0 \tag{65}$$

For covariant vectors:

$$[\nabla_\mu, \nabla_\nu]V_\rho = -R_{\rho\sigma\mu\nu}V^\sigma \tag{66}$$

Note: For Minkowski spacetime, $\Gamma^\mu_{\alpha\beta} = 0$ everywhere, so $R^\rho_{\sigma\mu\nu} = 0$.

5.3 Ricci Tensor and Ricci Scalar

Definition 1 (Ricci Tensor)**.** *The Ricci tensor is obtained by contracting the first and third indices:*

$$R_{\mu\nu} = R^\rho_{\mu\rho\nu} = g^{\alpha\beta}R_{\alpha\mu\beta\nu} \tag{67}$$

Definition 2 (Ricci Scalar)**.** *The Ricci scalar (or scalar curvature) is:*

$$R = g^{\mu\nu}R_{\mu\nu} \tag{68}$$

The Ricci scalar is a scalar quantity with at most two derivatives of the metric. It plays a central role in the Einstein-Hilbert action.

6 The Einstein-Hilbert Action

6.1 Requirements for Gravitational Action

The action for the metric $g_{\mu\nu}$ must satisfy:

1. It should be a scalar under general coordinate transformations
2. It should contain at most 2 derivatives of $g_{\mu\nu}$ (to match Maxwell's theory)
3. It should reduce to Newtonian gravity in the appropriate limit

General form:

$$S_G = \int d^4x\sqrt{-g}\, f(g_{\mu\nu}, \partial_\alpha g_{\mu\nu}, \partial_\alpha\partial_\beta g_{\mu\nu}) \tag{69}$$

The function f must be a scalar. The only scalar with at most two derivatives is the Ricci scalar R (and constants).

6.2 Dimensional Analysis

The action has dimensions:

$$[S] = [ML^2T^{-1}] \tag{70}$$

The metric is dimensionless: $[g_{\mu\nu}] = 1$
Newton's constant: $[G] = M^{-1}L^3T^{-2}$
The Ricci scalar: $[R] = L^{-2}$
For the action to have correct dimensions:

$$[\alpha] = \frac{[S]}{L^4[R]} = \frac{ML^2T^{-1}}{L^4 \cdot L^{-2}} = \frac{ML^2T^{-1}}{L^2} = MT^{-1} = \frac{c^3}{G} \tag{71}$$

6.3 The Einstein-Hilbert Action

Including a cosmological constant term:

$$\boxed{S_{EH} = \frac{c^3}{16\pi G}\int d^4x\sqrt{-g}\,(R - 2\lambda)} \tag{72}$$

where:

- λ is the **cosmological constant**
- The factor 16π is chosen to reproduce Newtonian gravity in the weak-field limit
- This action contains at most 2 derivatives of the metric

Setting $c = 1$ (natural units), we often write:

$$S_{EH} = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,R \tag{73}$$

Or with convenient normalization:

$$S_{EH} = \frac{1}{2}\int d^4x\sqrt{-g}\,R \tag{74}$$

7 Scalar Field Theory in Curved Spacetime

7.1 Action for Minimally Coupled Scalar Field

7.1.1 Flat Spacetime Action

In flat Minkowski spacetime, the action for a real scalar field is:

$$S_\phi^{\text{flat}} = \int d^4x \left[\frac{1}{2}\partial^\mu\phi\partial_\mu\phi - V(\phi)\right] \tag{75}$$

Requirements for the Lagrangian density:

- $\mathcal{L}$ must be a scalar
- At most two derivatives in equations of motion
- No explicit coordinate dependence (translational invariance)

General form:

$$\mathcal{L} = A(\phi)\partial_\mu\phi\partial^\mu\phi - U(\phi) \tag{76}$$

By field redefinition $\chi(\phi) = \int^\phi d\varphi\sqrt{2A(\varphi)}$, this can always be brought to canonical form:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\chi\partial^\mu\chi - V(\chi) \tag{77}$$

7.1.2 Curved Spacetime Action

To promote the scalar field action to curved spacetime (minimal coupling):

- $\partial^\mu\phi\partial_\mu\phi \to g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$
- $d^4x \to \sqrt{-g}\, d^4x$
- $\partial_\mu \to \nabla_\mu$ (though for scalars this makes no difference)

The curved spacetime action becomes:

$$\boxed{S_\phi = \int d^4x\sqrt{-g}\left[\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right]} \tag{78}$$

7.2 Equations of Motion

Varying with respect to ϕ and setting $\delta S = 0$:

$$\delta S_\phi = \int d^4x\sqrt{-g}\left[g^{\mu\nu}(\partial_\mu\phi)\partial_\nu(\delta\phi) - V_{,\phi}\delta\phi\right] \tag{79}$$

Integrating by parts and requiring $\delta S_\phi = 0$ for arbitrary $\delta\phi$:

$$\boxed{\Box_g\phi + V_{,\phi} = 0} \tag{80}$$

where the curved spacetime d'Alembertian operator is:

$$\Box_g = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\, g^{\mu\nu}\partial_\nu\right) \tag{81}$$

This is the **covariant Klein-Gordon equation** in curved spacetime.

7.3 Full Action: Gravity + Matter

The complete action coupling gravity to a scalar field:

$$\boxed{S = S_{EH} + S_\phi = \int d^4x\sqrt{-g}\left[\frac{1}{2}R + \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right]} \tag{82}$$

8 Scalar Field Cosmology

8.1 FRW Metric

The Friedmann-Robertson-Walker (FRW) metric for a homogeneous and isotropic universe:

$$ds^2 = -dt^2 + a^2(t)\left[dr^2 + r^2 d\Omega^2\right] = -dt^2 + a^2(t)d\vec{x}^2 \tag{83}$$

The metric components:

$$g_{\mu\nu} = \mathrm{diag}(-1, a^2, a^2, a^2) \tag{84}$$

The determinant:

$$g = \det(g_{\mu\nu}) = -a^6 \quad \Rightarrow \quad \sqrt{-g} = a^3 \tag{85}$$

8.2 Friedmann-Klein-Gordon Equation

For a homogeneous scalar field $\phi = \phi(t)$:

$$g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = -\dot{\phi}^2 \tag{86}$$

Computing the d'Alembertian:

$$\Box_g\phi = \frac{1}{a^3}\partial_t(a^3 \times (-\dot{\phi})) = -\ddot{\phi} - 3\frac{\dot{a}}{a}\dot{\phi} \tag{87}$$

The Klein-Gordon equation $\Box_g\phi + V_{,\phi} = 0$ becomes:

$$\boxed{\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0} \tag{88}$$

where $H = \dot{a}/a$ is the Hubble parameter.
Physical interpretation:

- $\ddot{\phi}$: acceleration of the field
- $3H\dot{\phi}$: Hubble friction (damping due to cosmological expansion)
- $V_{,\phi}$: restoring force from potential gradient

8.3 Stress-Energy Tensor for Scalar Field

The stress-energy tensor is obtained by varying the matter action with respect to the metric:

$$T_{\mu\nu} = \frac{-2}{\sqrt{-g}}\frac{\delta S_\phi}{\delta g^{\mu\nu}} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left(\frac{1}{2}\partial^\sigma\phi\partial_\sigma\phi - V(\phi)\right) \tag{89}$$

For a homogeneous field in FRW cosmology, this takes the form of a perfect fluid:
Energy density:

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi) \tag{90}$$

Pressure:

$$P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi) \tag{91}$$

Equation of state parameter:

$$w_\phi = \frac{P_\phi}{\rho_\phi} = \frac{\frac{1}{2}\dot{\phi}^2 - V}{\frac{1}{2}\dot{\phi}^2 + V} \tag{92}$$

8.4 Inflationary Scenarios

When the potential energy dominates ($V \gg \frac{1}{2}\dot{\phi}^2$):

- $w_\phi < 0$: negative pressure
- $w_\phi < -\frac{1}{3}$: accelerated expansion

This provides a mechanism for cosmological inflation.

8.5 Friedmann Equations

The dynamics of the scalar field coupled to gravity are governed by:

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0 \quad \text{(Klein-Gordon)} \tag{93}$$

$$H^2 = \frac{1}{3}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right) \quad \text{(Friedmann, assuming } k = 0\text{)} \tag{94}$$

These equations describe the evolution of a homogeneous scalar field in an expanding FRW universe (flat spatial sections, $k = 0$).

9 Maxwell's Equations in Curved Spacetime

9.1 Covariant Form

The Maxwell equations in curved spacetime can be written using the covariant divergence:

$$\boxed{\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu} \tag{95}$$

where the covariant divergence is:

$$\nabla_\mu F^{\mu\nu} = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\, F^{\mu\nu}\right) \tag{96}$$

Explicitly:

$$\partial_\mu\left(\sqrt{-g}\, F^{\mu\nu}\right) = \sqrt{-g}\, \mu_0 J^\nu \tag{97}$$

In flat spacetime ($\sqrt{-g} = 1$), this reduces to:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu \tag{98}$$

9.2 Important Note on Field Tensor

The electromagnetic field tensor is defined as:

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{99}$$

The Christoffel symbols cancel in the antisymmetric combination, so the covariant and ordinary derivatives give the same result for $F_{\mu\nu}$.

10 Summary of Key Results

10.1 Special Relativity

- Lorentz transformations with $\gamma = (1 - v^2/c^2)^{-1/2}$
- Four-vectors: position, velocity, momentum, force
- Electromagnetic field tensor $F^{\mu\nu}$ and covariant Maxwell equations
- Gauge invariance and electromagnetic action

10.2 General Relativity

- Invariant volume element: $dV = \sqrt{-g}\, d^4x$
- Covariant derivative with Christoffel symbols
- Metric compatibility: $\nabla_\mu g_{\alpha\beta} = 0$
- Riemann curvature tensor measures spacetime curvature
- Einstein-Hilbert action: $S_{EH} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g}(R - 2\lambda)$

10.3 Scalar Field Cosmology

- Minimally coupled scalar field action in curved spacetime
- Covariant Klein-Gordon equation
- Friedmann-Klein-Gordon equation for homogeneous fields
- Scalar field as perfect fluid with equation of state parameter w_ϕ
- Mechanism for inflation when potential dominates